Knowledge Graphs

Your Portfolio

Emanuel Sallinger

How am I going to be assessed?

By one item: the mini-project (“portfolio”):

1. **one-page summary** proposing a mini-project by you
2. **feedback** on the proposal
3. submission of the **final mini-project report** ("portfolio")

The aim of this is course is to gain a deep understanding of Knowledge Graphs divided into three blocks. An overarching aim of the course is to understand the connections between Knowledge Graphs, Artificial Intelligence, Machine Learning and Data Science.

Representations

Logical and ML-based representations for KGs.

Read more details

KG Embeddings
Widely-applied, large family of ML models.

Logical Knowledge in KGs
Highly expressive, diverse family of logical models.

Graph Neural Networks
Using the KG structure *as* a neural network.

Data Models
Overview of data models in different communities.

Systems

Systems to bring KGs into practice.

Read more details

Architectures
The big picture of building IT architectures for KGs.

Scalable Reasoning
Making use of the knowledge in the KG.

KG Creation
How to create a KG from heterogeneous data?

KG Evolution
How to update, correct and complete a KG?

Applications

Real-world applications of KGs.

Read more details

Real-World Applications
Overview of diverse applications.

Financial KGs
Concrete applications in finance and economics.

Services
Which service to provide based on KGs?

Connections
... between KGs, AI, ML and Data Science.

https://kg.dbai.tuwien.ac.at/kg-course/

COURSE DELIVERY

How Am I Going to be Assessed?

Grading is according to the following principles:

- G4: Showing basic proficiency in at least 6 learning outcomes.
- B3: Showing basic proficiency in at least 10 learning outcomes.
- U2: As above, and exceeding the threshold in at least 1 learning outcome.
- S1: As above, and exceeding the threshold in at least 2 learning outcomes.

That is, it is not necessary for each portfolio to go beyond basic proficiency in all learning outcomes, but perfectly fine to excel in a selection of them.

https://kg.dbai.tuwien.ac.at/kg-course/

COURSE DELIVERY

How Am I Going to be Assessed?

Summative assessment is by one item: the project **portfolio**. It was already briefly mentioned in the overview. In particular, it:

- gives a *clear, transparent basis* for the achieved marks, and puts control of demonstrating the achieved learning outcomes into one's own hands.
- allows *diversity* in how to demonstrate the learning outcomes: while a typical way to demonstrate that would be a practical project portfolio, those who are pursuing a theoretical direction will be able to demonstrate that through a theoretical project portfolio
- allows for *self-regulated learning*: while the portfolio is the single assessment item, it is the witness of a longer learning process, and can be created - using the principle of "patchwork" text - throughout the learning process, or in one block at the end, depending on the learning type

COURSE DELIVERY

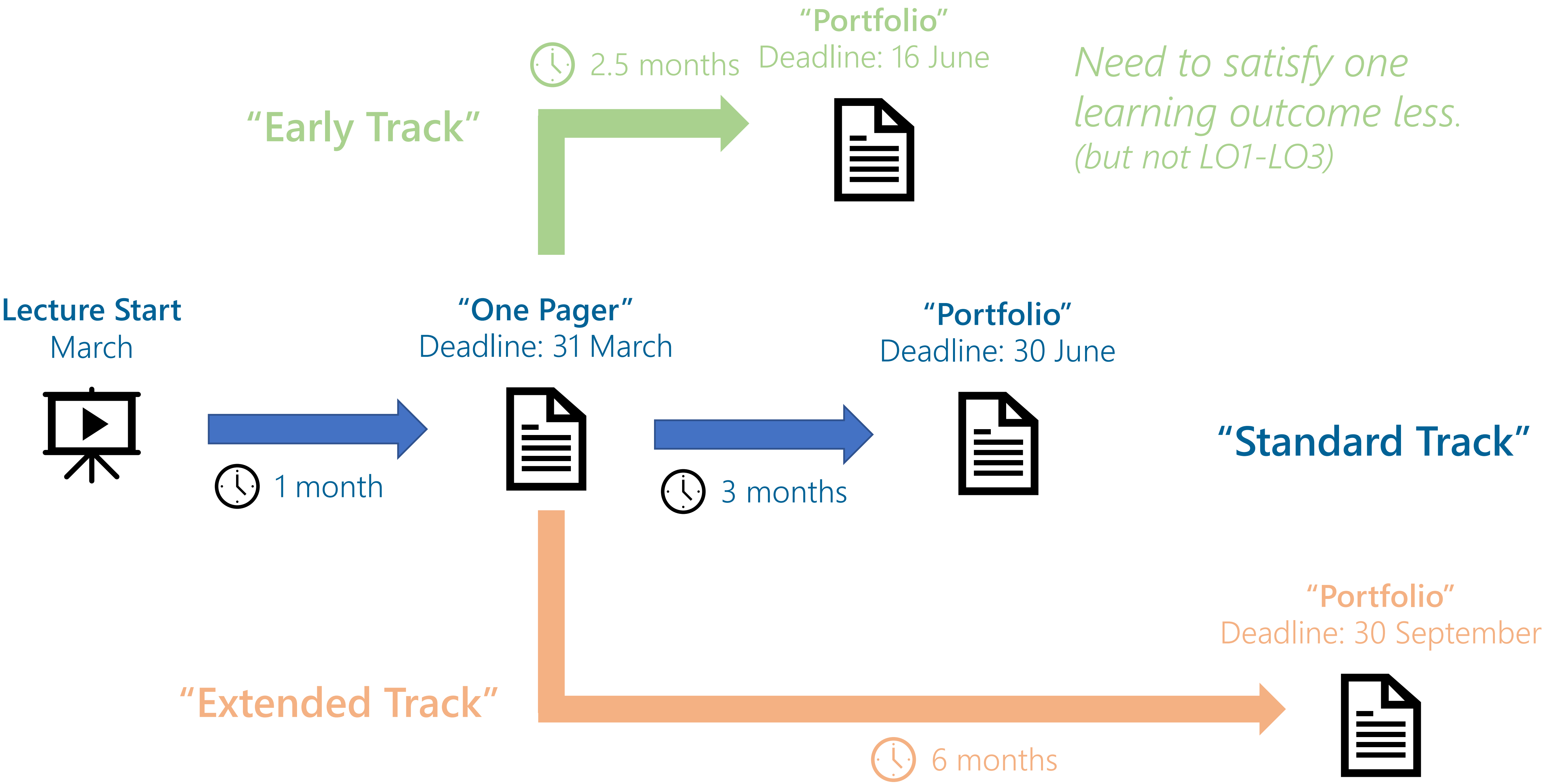
How Can a Portfolio Look Like?

The following is a **(non-exhaustive) list of examples** of what can make a good project and portfolio topic:

- **Application-oriented**: Portfolio on creating a financial Knowledge Graph, based on public data in Austria combined with EU-level open data.
- **System-oriented**: Portfolio on a KG used for scalable reasoning using one ML-technique and a logic-based technique.
- **Foundations-oriented**: Portfolio on complexity of KG processing and reasoning in an, e.g., databases or ML-based KG framework.
- **State-of-the-art-oriented**: Portfolio on the state-of-the-art of Graph Neural Network-based reasoning that incorporates domain knowledge for a chosen domain, and shows how the techniques could be applied.

Each portfolio will have a particular focus, but needs to put it in the context of the other learning outcomes. E.g., while not every portfolio will apply KG Embeddings based models, it should put the covered topics in the context, to demonstrate meeting the learning outcomes.

https://kg.dbai.tuwien.ac.at/kg-course/


"Portfolio"
Deadline: 16 June
2.5 months
"Early Track"
Need to satisfy one learning outcome less.
(but not LO1-LO3)
Lecture Start
March
1 month
"One Pager"
Deadline: 31 March
3 months
"Portfolio"
Deadline: 30 June
"Standard Track"
"Extended Track"
6 months
"Portfolio"
Deadline: 30 September

Knowledge Graphs

Your Portfolio

Emanuel Sallinger